

Adrien Goldszal

curriculum vitae

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in [adrien-goldszal](#)

Education

- 2025-2026 **Master Year 2 : AI, Systems & Data**, PSL Research University, Paris, France
Theoretical foundations of machine learning.
Relevant coursework includes : *Reinforcement Learning, LLMs, Optimal Transport*
- 2022-2026 **Diplôme d'ingénieur**, Ecole Polytechnique, Palaiseau, France
Major in Advanced Science and Engineering, focusing on computer science & applied mathematics.
Third year specialisation in artificial intelligence and data science.
Relevant coursework includes : *Optimisation and Control, Machine Learning and Autonomous Agents, Image Analysis and Computer Vision, Reliable Collaborative Learning*
GPA 3.88/4.0
- 2020-2022 **French Preparatory Classes**, Lycée Privé Sainte Geneviève, Versailles, France
Intensive two-year preparation in mathematics, physics and chemistry for the entrance exams of France's best *Grandes Ecoles*.
Undergraduate coursework includes *Linear Algebra, Real Analysis, Solid and Fluid Mechanics*
GPA 3.89/4.0
- 2017-2020 **Baccalauréat - Abitur**, Lycée International, Saint-Germain-En-Laye, France
Double French and German Baccalauréat and Abitur.
19.27/20 (1.0) Highest Honours

Work Experience

- April-Aug. **AI Research Intern**, Mila - Quebec AI Institute
2025 Research internship under the supervision of Dr. Pierre-Luc Bacon on generative modeling for molecule discovery : *"Discovery of Sustainable Refrigerants through Physics-Informed RL Fine-Tuning of Sequence Models"*
- June-Aug. **AI Research Intern**, Siemens Technology, Munich, Germany
2024 Three-month internship in the Digital Manufacturing Technologies research group under the supervision of Dr. Peter Robl
Deep reinforcement learning (RL) for automated electrical cable routing in continuous 3D space.
 - Implemented a search heuristic and leveraged demonstrations to improve the RL frameworks' performance
 - Used Explainable AI techniques like Shapley Values to optimize agent observations
 - Developed an API for Siemens clients facilitating the use of the lab's cable routing tools
- 2022-2023 **Officer Cadet & Assistant Project Manager**, Southeastern France Regional Fire Department, Bourg-en-Bresse, France
Ecole Polytechnique first year military internship (Six months).
 - Worked as a first responder, performing first aid & vehicle extractions.
 - Developed new and optimized tools to monitor and regulate the service's energy consumption
 - Coordinated a plan to reduce the service's carbon emissions by 30% (60 fire stations, 4000+ people)

Selected Research & Project Work

Course-related : Computer Science

- Feb.-March 2025 **Self-play for learning adversarial board games**, *École Polytechnique*
Developed a PettingZoo environment for the board game Cathedral and implemented and benchmarked algorithms for adversarial self-play learning for Polytechnique's *Autonomous Agents* course.
- Feb.-March 2025 **Seamless Dual Style Image Generation with Diffusion Models**, *Ecole Polytechnique*
Modified the Stable Diffusion architecture to smoothly interpolate between two prompts for a seamless "double style transfer" on a single output image, leveraging the cross-attention layers and the U-Net output
- Nov.-Dec. 2024 **Cooperative Inverse Reinforcement Learning (CIRL)**, *Ecole Polytechnique*
Conducted an in-depth analysis and implementation of two research papers on CIRL in the specific case of apprenticeship learning (Dylan Hadfield-Menell et al. Neurips 2016), and in the general CIRL framework (Dhruv Malik et al. ICML 2018) for Ecole Polytechnique's *Collaborative & Reliable Learning* course.
- Nov.-Dec. 2024 **Automatic Puzzle Solver**, *Ecole Polytechnique*
Two-month project for Polytechnique's *Image Analysis and Computer Vision* course.
 - Developed an automated real-time puzzle-solving system leveraging SIFT-based feature detection, edge detection, and morphological operations for robust piece identification
 - Enhanced accuracy in puzzle solving through manual calculation of homography transformations and validating placements using Zero-mean Normalized Cross-Correlation (ZNCC)
- Feb.-March 2024 **3D Animated Scene**, *Ecole Polytechnique*
Two-month project for Polytechnique's Computer Graphics Course.
 - Created a 3D animated scene in an open environment using the OpenGL library.
 - Modeled and integrated real-time physics, resulting in a more realistic and immersive user experience and implemented user commands for interactive control.

Course-related : Engineering

- 2023–2024 **Atmospheric Balloon**, *Ecole Polytechnique Spatial Center*
One year group engineering research project in collaboration with the LATMOS (French National Meteorology Lab).
 - Created new adapted tools and sensors sent on an atmospheric balloon for the study of multiple physical phenomena including conductivity and the electrical field of the earth
 - Developed the code for efficient data acquisition, transmission and analysis.
- 2021–2022 **Study of sand dune movement**, *Lycée Sainte-Geneviève*
One year group scientific research project for the entrance examinations of the French *Grandes Ecoles*.
 - Designed and built an experimental water tank system with automated sand displacement mechanisms to simulate desert dune formation at 1:1000 scale.
 - Implemented a Python-based cellular automaton model incorporating fluid dynamics principles, successfully predicting dune migration patterns.

Publications

Preprints

- Aug. 2025 **Discovery of Sustainable Refrigerants through Physics-Informed RL Fine-Tuning of Sequence Models**, *Arxiv* <https://arxiv.org/abs/2509.19588>

Activities & Volunteering

2023–Present **President & Software Team Leader, IP Paris Racing Team**

IP Paris Racing Team is an association created in 2023 to represent the Institut Polytechnique de Paris at Formula Student: designing an electric car to participate in international student competitions.

- Coordinating a team of 50 students, raising funds and finding sponsors for IP-Paris' first Formula Student vehicle
- Leading a team of 7 on the electronics of the car and the associated software. Working on the code for the communication between sensors in embedded C.

2024–Present **Core Member, X – AI (Polytechnique AI Association)**

X-AI helps promote and motivate students wanting to discover more about the AI sector.

- Organized multiple hackathons and conferences with leading AI research firms such as Google DeepMind & QuantumBlack
- Held introductory courses and tutoring sessions for first- and second-year students

Languages

French	Native	<i>French Baccalauréat (2020)</i>
German	C2 - Proficient	<i>German Abitur (2020)</i>
English	C2 - Proficient	<i>Cambridge Proficiency Exam : Grade A (2020)</i>
Spanish	B2 - Intermediate	

Technical Skills

ML/AI Frameworks	PyTorch, TensorFlow, Scikit-learn, OpenCV, OpenAI Gym
Data Engineering	SQL, Spark, Pandas, NumPy
Programming	Python, C++, Java, Git
Hardware	PCB Design, Arduino, Sensors